

Lecture FYS5419, April 10, 2024

FYS 5419/9419, APRIL 10, 2024

$$|j\rangle = |j_0, j_1, \dots, j_{n-1}\rangle$$

Example 2 qubits

$$|00\rangle \rightarrow |0\rangle \quad |j_0=0, j_1=0\rangle$$

$$|01\rangle \rightarrow |1\rangle \quad = |0\rangle \otimes |1\rangle$$

$$|10\rangle \rightarrow |2\rangle$$

$$|11\rangle \rightarrow |3\rangle$$

3-qubits

$$|000\rangle = |0\rangle \otimes |0\rangle \otimes |0\rangle \Rightarrow |0\rangle$$



$$J = J_0 z^{n-1} + J_1 z^{n-2} + \dots + J_{n-2} z^1 + J_{n-1} \cdot z^0$$

$$|J\rangle = \frac{1}{\sqrt{2^n}} \bigotimes_{m=0}^{n-1} \left\{ |0\rangle + e^{i2\pi J z^{-m}} |1\rangle \right\}$$

$$= \frac{1}{\sqrt{2^n}} \left[\begin{aligned} & (|0\rangle + e^{i2\pi(J/2^0)} |1\rangle) \otimes \\ & (|0\rangle + e^{i2\pi(J/2^1)} |1\rangle) \otimes \\ & \dots \otimes \dots \otimes (|0\rangle + e^{i2\pi(J/2^{n-1})} |1\rangle) \end{aligned} \right]$$

$$0. J_m J_{m+1} \dots J_{n-1} = J_m z^{-\frac{1}{2}} +$$

$$J_{m+1} 2^{-2} + \dots + J_{n-1} 2^{-(n-m)}$$

$$n > m$$

$$\begin{aligned} |J\rangle &= |J_0 J_1 J_2 \dots J_{n-1}\rangle \\ &= \frac{1}{\sqrt{2^n}} \left(|0\rangle + e^{i2\pi 0 \cdot J_{n-1}} |1\rangle \right) \otimes \\ &\quad \left(|0\rangle + e^{i2\pi 0 \cdot J_{n-2} J_{n-1}} |1\rangle \right) \otimes \\ &\quad \dots \otimes \left(|0\rangle + e^{i2\pi 0 \cdot J_0 J_1 J_2 \dots J_{n-1}} |1\rangle \right) \end{aligned}$$

$$\frac{J_0}{2} = 0 \cdot J_0 \quad 1 \cdot 0 \cdot J_0 J_1 = J_0 \frac{1}{2} + J_1 \frac{1}{2^2}$$

$0, J_0, J_1, J_2, \dots, J_{n-1}$ or

$0, J_{n-1}, J_{n-2}, \dots, J_1, J_0$

Rotations

$$e^{i\phi} = \cos \phi + i \sin \phi$$

$$e^{i\pi/2} = i \Rightarrow 90^\circ \text{ ccw}$$

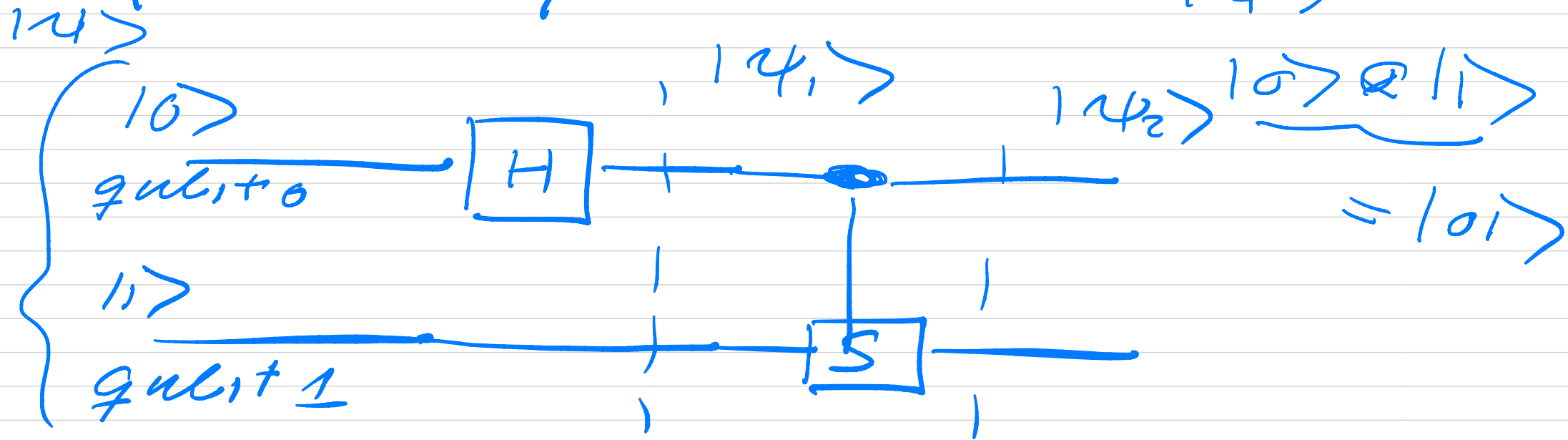
$$e^{i\pi} = -1 \Rightarrow 180^\circ \text{ ccw}$$

$$e^{i3\pi/2} = -i \Rightarrow 270^\circ \text{ ccw}$$

90° cw



QFT 2 qubits



$$|14_1\rangle = H_{(2 \times 2)} \otimes I_{(2 \times 2)} |01\rangle$$

$$= \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \otimes |1\rangle$$

$$|14_2\rangle = \frac{1}{\sqrt{2}} (|0\rangle \otimes |1\rangle + |1\rangle \otimes |1\rangle)$$

$$S|1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/2} \end{bmatrix} |1\rangle$$

$$= e^{i\pi/2} |1\rangle = \Rightarrow$$

$$|4_2\rangle = \frac{1}{\sqrt{2}} \left[|10\rangle \otimes |1\rangle + e^{i\pi/2} |1\rangle \otimes |1\rangle \right]$$

$$= \frac{1}{\sqrt{2}} \left[|100\rangle + e^{i\pi/2} |11\rangle \right]$$

Rotation

$$R_k = \begin{bmatrix} 1 & 0 \\ 0 & e^{2\pi i / 2^k} \end{bmatrix}$$

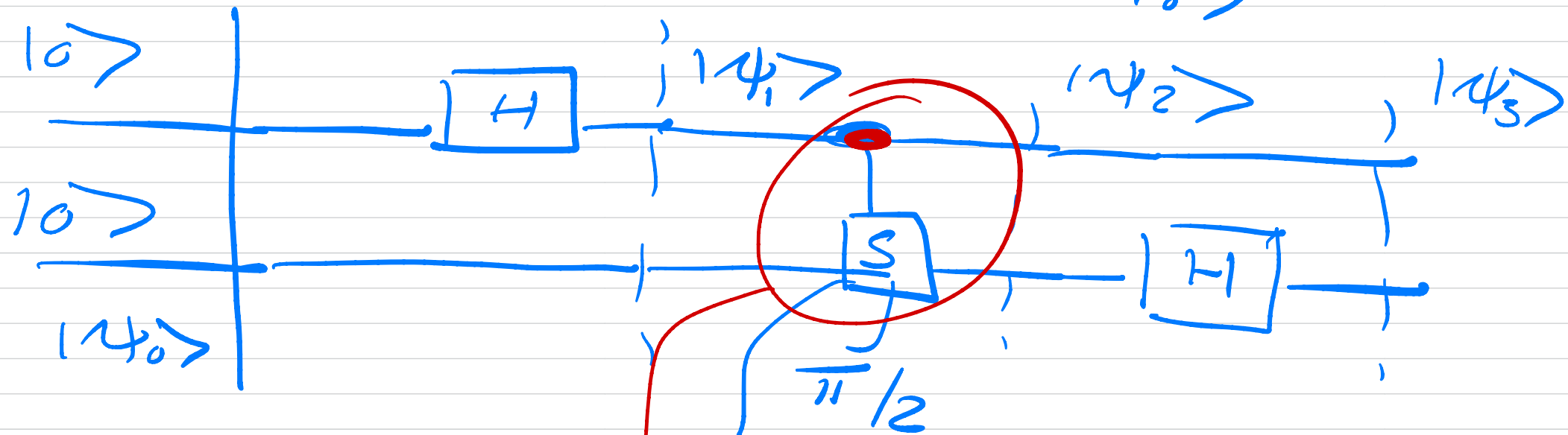
$$R_{k=0} = \begin{bmatrix} 1 & 0 \\ 0 & e^{2\pi i} \end{bmatrix}$$

$$R_{k=1} = \begin{bmatrix} 1 & 0 \\ 0 & e^{2\pi i / 2} \end{bmatrix}$$

FULL QFT FOR 2 qubits

start with $|00\rangle = |0\rangle \otimes |0\rangle$

$$= |4_0\rangle$$



control-R gate

$$|4_0\rangle = |0\rangle \otimes |0\rangle = |00\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} |4_1\rangle &= (H \otimes I) |00\rangle \\ &= \frac{|00\rangle + |10\rangle}{\sqrt{2}} \end{aligned}$$

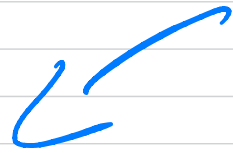
$$|4_2\rangle = |4_1\rangle$$

$$\begin{aligned} |4_3\rangle &= (I \otimes H) |4_2\rangle \\ &= \frac{1}{2} [|00\rangle + |01\rangle + |10\rangle + |11\rangle] \end{aligned}$$

$$|4_0\rangle = |01\rangle$$

$$|4_3\rangle = \frac{1}{2} (|00\rangle - |01\rangle + i|10\rangle - i|11\rangle)$$

$$|0\rangle \rightarrow \text{QFT } |j\rangle$$



$$\frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & i & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -i & i \end{bmatrix}$$

3 qubits

$$R_1 = \begin{bmatrix} 1 & 0 \\ 0 & e^{i2\pi/2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & w \end{bmatrix}$$

$$w =$$

$$w = e^{2\pi i / 2^n}$$

$$w = e^{i\pi}$$

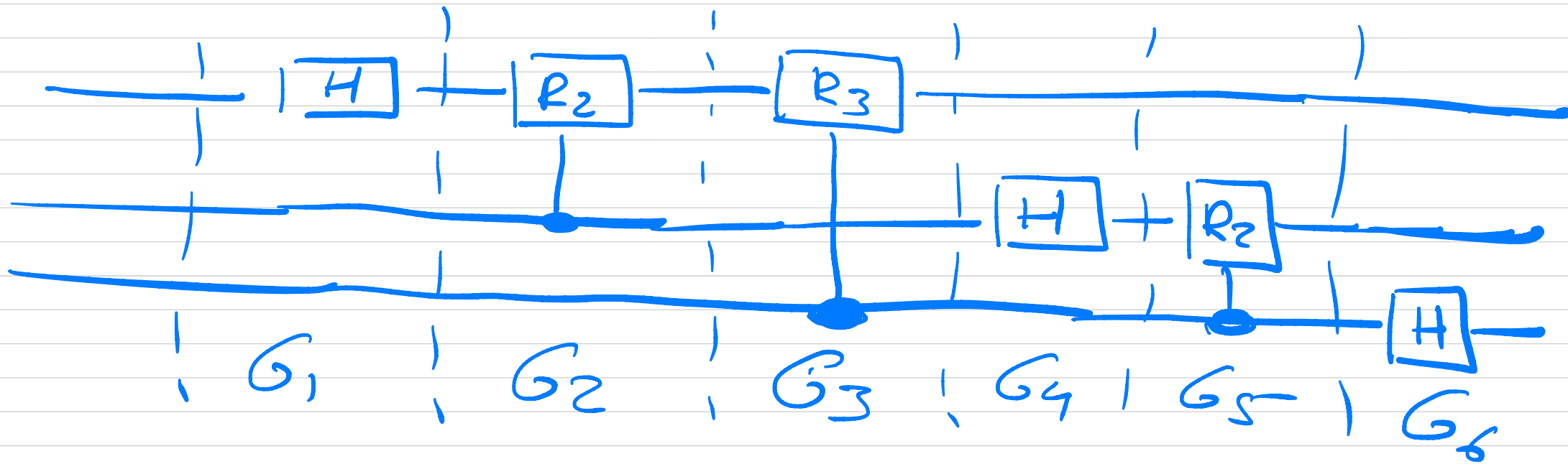
$$R_2 = \begin{bmatrix} 1 & 0 \\ 0 & e^{i2\pi/2^2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & \omega^2 \end{bmatrix}$$

$$R_3 = \begin{bmatrix} 1 & 0 \\ 0 & e^{i2\pi/2^3} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & \omega^3 \end{bmatrix}$$

Control - R_2 gate is

$$\begin{bmatrix} 1 & & 0 \\ & 1 & \\ 0 & & e^{i2\pi/2^2} \end{bmatrix} = \begin{bmatrix} 1 & & 0 \\ & 1 & \\ C & & \omega^2 \end{bmatrix}$$

Circuit is



$$G_I = H \otimes I \otimes I$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} I_4 & I_4 \\ I_4 & -I_4 \end{bmatrix} \quad I_4 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$G_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & \omega^2 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ c & 1 \end{bmatrix}$$

$$= \text{Control-}R_2 \otimes I$$

$$G_2 = \begin{bmatrix} I_4 & 0 \\ 0 & M_4 \end{bmatrix}$$

$$M_4 = \begin{bmatrix} 1 & 0 \\ 0 & \omega^2 \\ 0 & \omega^2 \end{bmatrix}$$

$$G_3 = \begin{bmatrix} I_4 & 0 \\ 0 & M_{43} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ u & 0 \\ 0 & u \end{bmatrix}$$

$$G_6 G_5 G_4 G_3 G_2 G_1 = QFT_3$$

==

$$\frac{1}{(\sqrt{2})^3}$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & \omega^1 & \omega^2 & \omega^3 & \omega^4 & \omega^5 & \omega^6 & \omega^7 \\ 1 & \omega^2 & \omega^4 & \omega^6 & 1 & \omega^2 & \omega^4 & \omega^6 \\ 1 & \omega^3 & \omega^6 & \omega & \omega^4 & \omega^7 & \omega^2 & \omega^5 \\ 1 & \omega^4 & 1 & \omega^4 & 1 & \omega^4 & 1 & \omega^4 \\ 1 & \omega^5 & \omega^2 & \omega^7 & \omega^4 & \omega^1 & \omega^6 & \omega^3 \\ 1 & \omega^6 & \omega^4 & \omega^2 & \omega^6 & \omega^4 & \omega^2 & 1 \\ 1 & \omega^7 & \omega^6 & \omega^5 & \omega^4 & \omega^3 & \omega^2 & \omega^1 \end{bmatrix}$$

$$|10, j_1, j_2\rangle = \left(\frac{1}{\sqrt{2}}\right)^3 (|10\rangle + e^{i2\pi 0 \cdot j_1 j_2} |11\rangle)$$

$$\otimes (|10\rangle + e^{i2\pi 0 \cdot j_1 j_2} |11\rangle) \otimes$$

$$Q(|0\rangle + e^{i2\pi C_0 J_0 J_1 J_2} |1\rangle)$$

QPE: QUANTUM PHASE
estimation. why?

unitary transformation
 U has an eigenvector

$$u_i$$

$$U = \begin{bmatrix} | & | & & | \\ u_0 & u_1 & \dots & u_{n-1} \\ | & & & | \end{bmatrix}$$

$$u_i^T u_j = \delta_{ij}$$

and eigenvalue

$$\lambda_i' = e^{2\pi i \phi_i'}$$

$$\phi_i' \Rightarrow \phi$$

$$\mathcal{U} u_i = \lambda_i u_i$$

$$\mathcal{U} |x\rangle = \lambda |x\rangle$$

$$\langle x | u^\dagger | u x \rangle = |\lambda|^2 \Rightarrow$$

$$|\lambda| = \underline{1}$$

$$\lambda = e^{2\pi i \phi}$$

$$\phi \in [0, 1]$$

Binary representation of ϕ
in terms of t -bits

$$\phi = 0.\phi_0\phi_1 \dots \phi_{t-1}$$

$$= \phi_0 \cdot \frac{1}{2^1} + \phi_1 \cdot \frac{1}{2^2} + \dots + \phi_{t-1} \cdot \frac{1}{2^t}$$

The t -bits represent the
precision we want for ϕ

$$t = \underline{1}$$

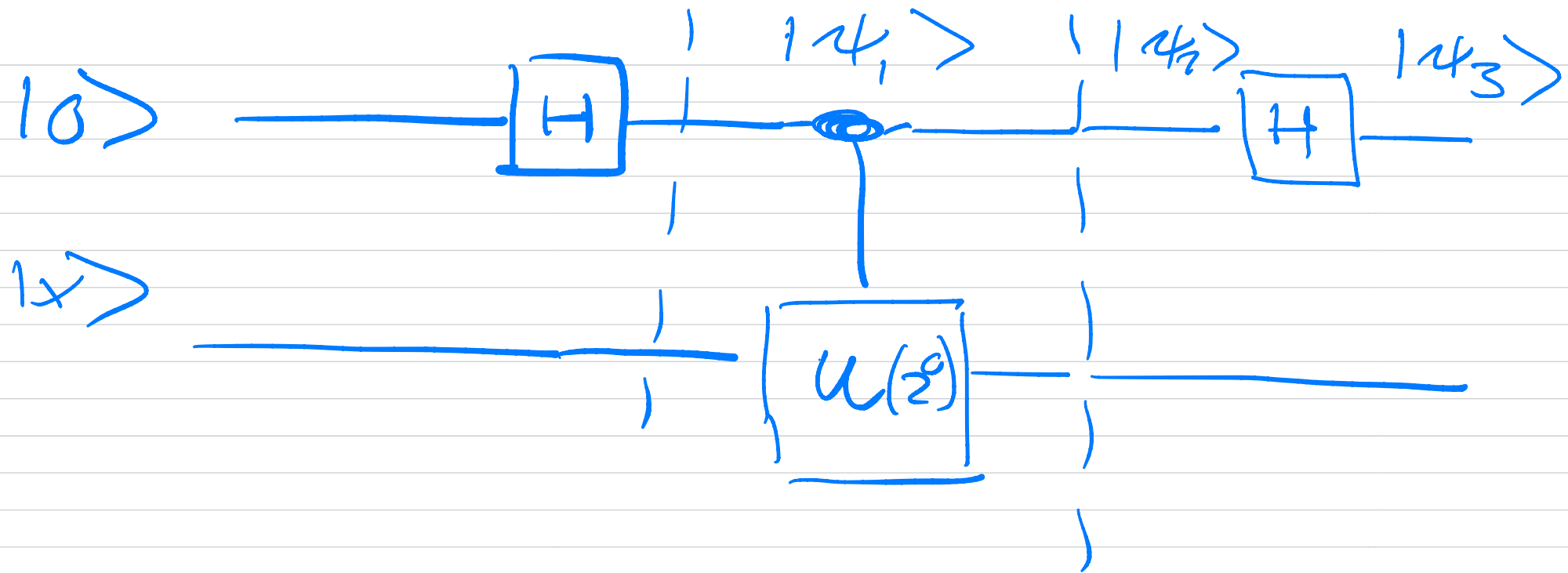
$$\lambda = e^{2\pi i \sigma \cdot \phi_0}$$

$$\phi_0 = \{0, 1\} \Rightarrow \phi = \{0, \frac{1}{2}\}$$

BASIC assumption:

initialize t -bits to $|0\rangle$
(First register)

The second register contains
all bits needed to define
the state $|x\rangle$



$$|\psi_1\rangle = \frac{1}{\sqrt{2}} (|0\rangle \otimes |x\rangle + |1\rangle \otimes |x\rangle)$$

control - U operation

$$|\psi_2\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle \otimes |x\rangle + e^{2\pi i 0 \cdot \phi_0} |1\rangle \otimes |x\rangle \right)$$

$$= \frac{1}{\sqrt{2}} (|0\rangle + e^{2\pi i \sigma_z \phi_0} |1\rangle) \otimes |x\rangle$$

Measuring gives either $|0\rangle$
or $|1\rangle$ with probability $\frac{1}{2}$

$$|\psi_3\rangle = H |\psi_2\rangle$$

$$= \frac{1}{2} (1 + e^{2\pi i \sigma_z \phi_0}) |0\rangle$$

$$+ (1 - e^{2\pi i \sigma_z \phi_0}) |1\rangle \otimes |x\rangle$$

ϕ_0 is either 0 or $\underline{1}$

$$\phi_0 = 0 \Rightarrow \text{then } |\psi_3\rangle = |0\rangle$$

$$\phi_0 = 1 \Rightarrow |\psi_3\rangle = |1\rangle$$

That means that if we measure either $|0\rangle$ or $|1\rangle$, with certainty, if

$\phi = 0$ or 0.5 , we know the eigenvalue since

$$\lambda = e^{2\pi i \phi}$$

Till next time, finalize
the DE algorithm in order
to find ϕ . Using QFT[†]

Hundt 6.1-6.4